RMSE-Anchored Contrastive Calibration for Reliable Long-Tail Regression on Urban Proxy Dynamics

Grid-level Prediction of Population and Nighttime Light Changes (2015–2024)

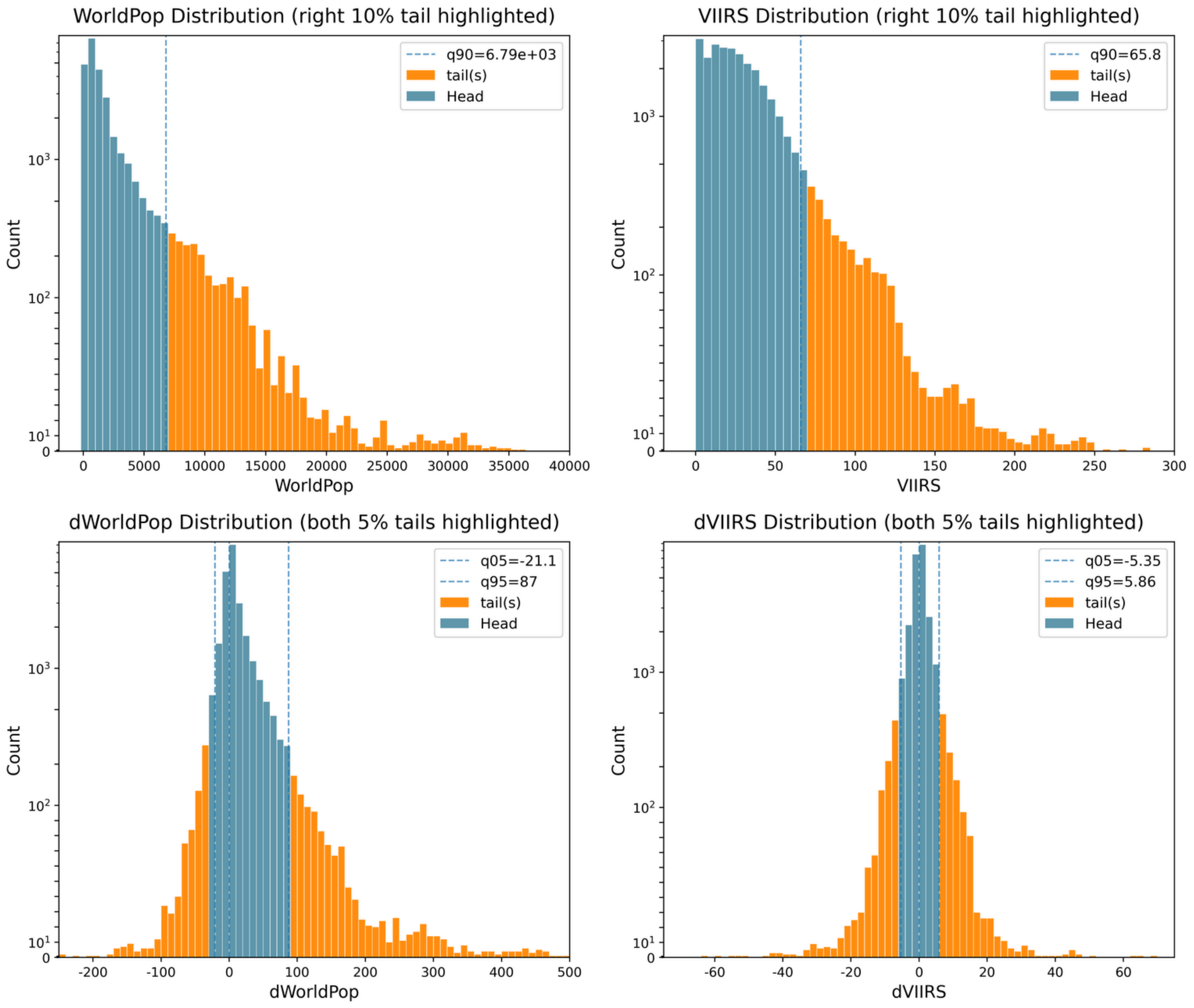
Winston HUANG

Research Question

Can Machine Learning model reliably predict economic dynamics via transportation accessibility-related features?

How can we reliably predict long-tailed urban proxy dynamics?

How could modeling strategies improve tail and worst-case performance?



Background

-Transportation infrastructure strongly influences urban economic outcomes, but direct economic statistics are costly, delayed, and spatially coarse.

-Scalable proxy signals such as population density (WorldPop) and nighttime lights (VIIRS) enable high-resolution urban economic analysis.

-Challenge:

Year-to-year proxy changes are heavy-tailed, causing standard regression models to underperform on rare but important shocks.

○

Literature Review

Long-Tail Regression

-Standard MSE/MAE biases models toward majority (head) samples

Prior work proposes:

Balanced MSE (Ren et al., 2022)

Label / Feature Distribution Smoothing (Yang et al., 2021)

Literature Review

Population as Proxy

-Population growth reflects economic capacity and productivity

-Supported by classical and modern economic studies

Nighttime Lights

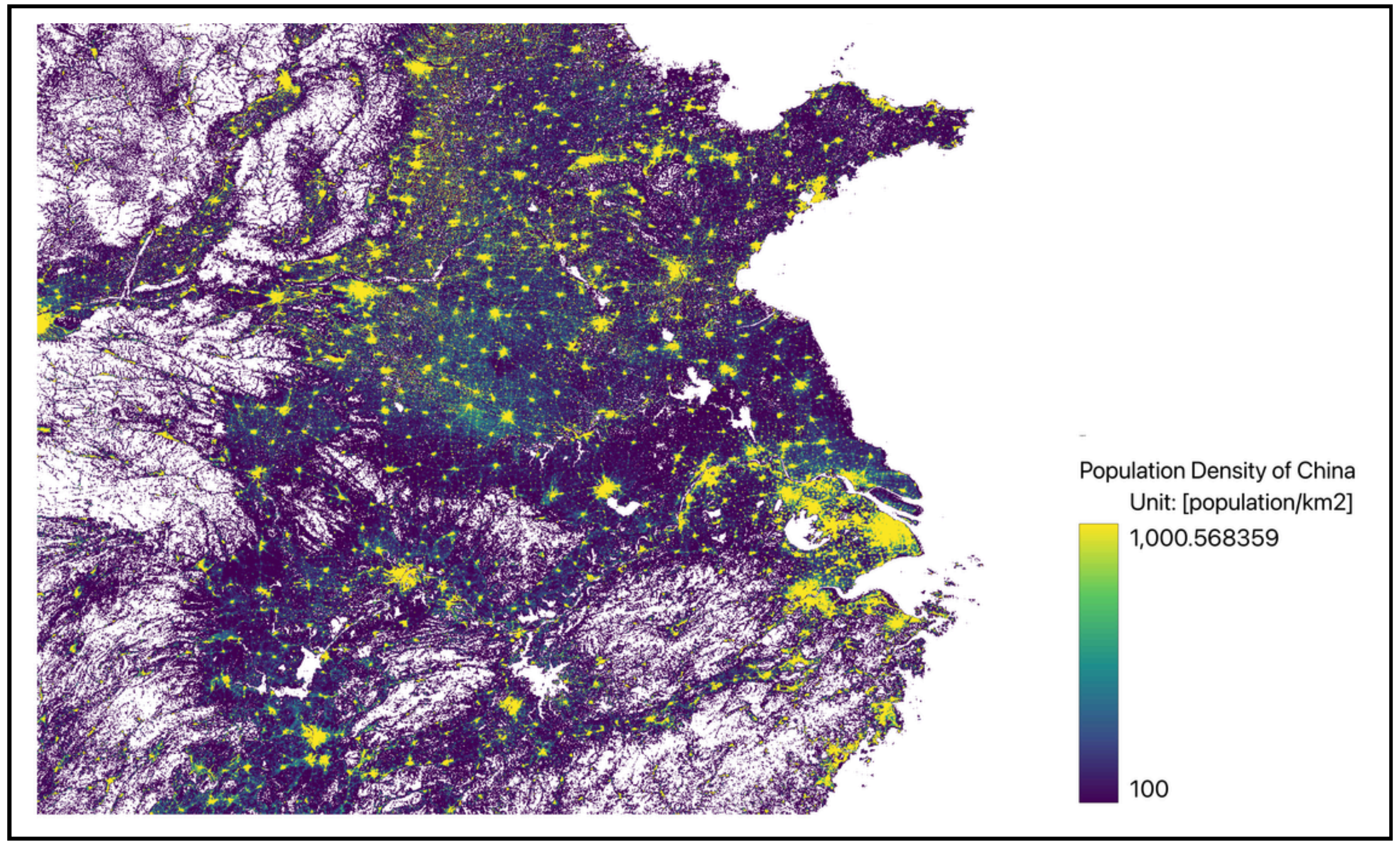
-Strong correlation with GDP at coarse scales

-Noisy at fine spatial and temporal resolution

Gap Identified

-Few benchmarks jointly address grid-level dynamics

+ long-tailed targets + robust evaluation



WorldPop, visualization by Winston Huang

Dataset: CivitasGrid-TLP (TransLuxPop)

Dataset Overview

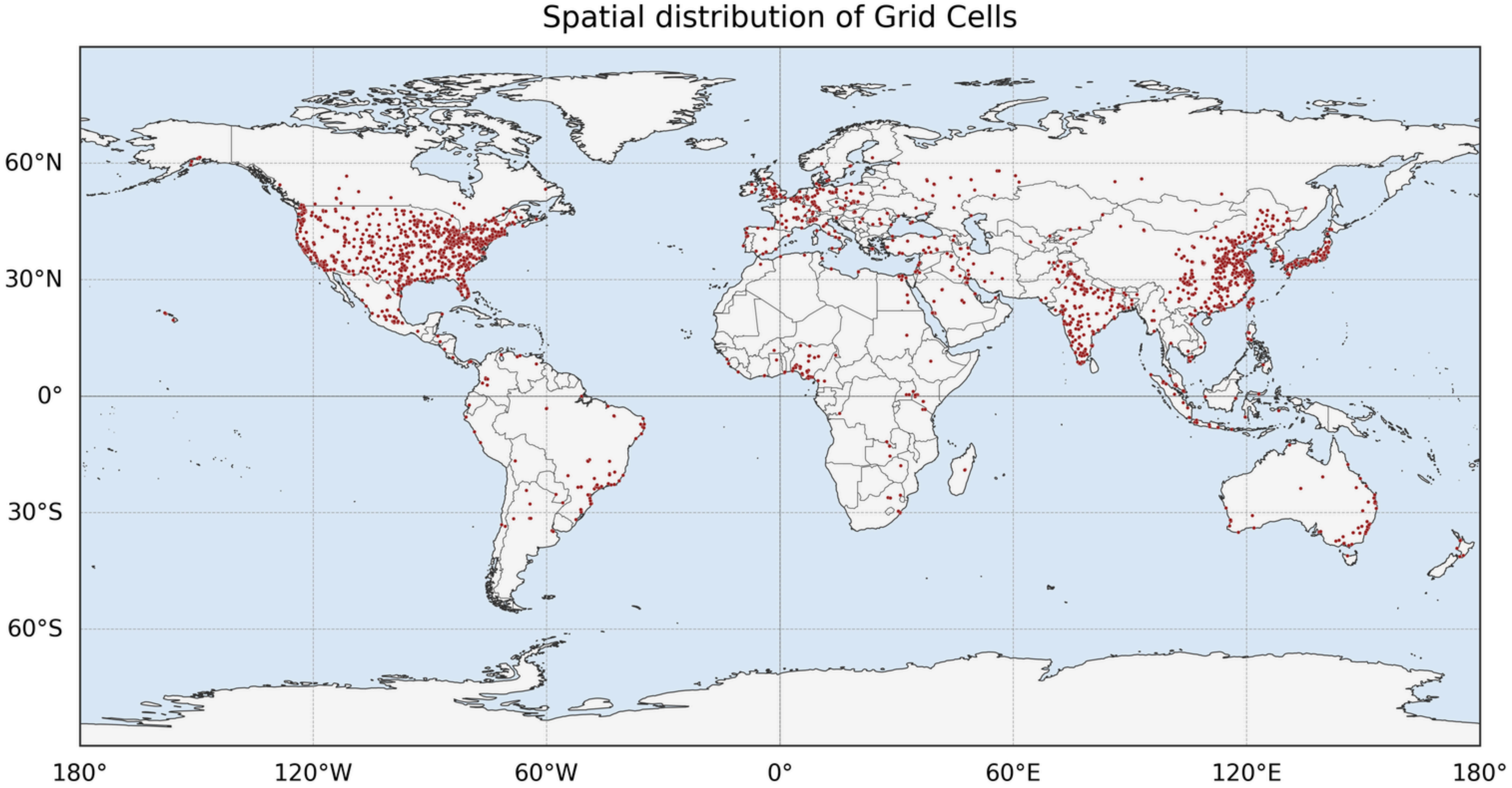
- Grid-level panel dataset (2015–2024)
- Global coverage, km-level resolution

Data Sources

- Transportation features: OpenStreetMap
- Proxy signals: WorldPop, VIIRS

Targets

- ΔWorldPop (population change)
- ΔVIIRS (nighttime light change)



Research Design: Problem Setup

Problem Setup & Tail Definition

- Tail samples = top 5% and bottom 5% of target distribution
- Focus on rare, high-magnitude shocks

Evaluation Focus

- Overall performance
- Tail-specific performance

$$i \in \mathcal{T}_{\text{VIIRS}} \iff \begin{cases} \Delta\text{VIIRS}_i \leq -5.35, \\ \Delta\text{VIIRS}_i \geq 5.86, \end{cases}$$

$$i \in \mathcal{T}_{\text{WorldPop}} \iff \begin{cases} \Delta\text{WorldPop}_i \leq -21.1, \\ \Delta\text{WorldPop}_i \geq 87. \end{cases}$$

Baseline Models & Contrastive Calibration

Contrastive learning evaluation

$$p_i = \frac{\exp(-|\tilde{y}_i - y_i|/T)}{\sum_{j \in \mathcal{B}} \exp(-|\tilde{y}_i - y_j|/T)},$$

Contrastive learning loss function

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} -\log p_i.$$

Mixed loss function

$$\mathcal{L}_{\text{mix}} = w_{\text{rmse}} \cdot \frac{\text{RMSE}}{\text{RMSE}_0} + w_{\text{cl}} \cdot \frac{\mathcal{L}_{\text{con}}}{\mathcal{L}_{\text{con},0}},$$

baseline model training

⇩

Calibration layer training

⇩

Results

- Without Calibration
- CL-only calibration
- RMSE-anchored calibration

Results

Benchmark Models

TABLE VI: Benchmark performance on ΔWorldPop (dWorldPop).

Split	Model	R^2	RMSE	R^2_{tail}	$RMSE_{tail}$
RandomSplit	XGBoost	0.836	21.976	0.832	61.048
	Ridge	0.354	43.593	0.366	118.483
	ElasticNet	0.274	46.197	0.215	131.789
	SVR	0.440	40.597	0.373	117.766
	ExtraTrees	0.843	21.499	0.826	62.033
	MLP	0.766	26.239	0.791	68.011
GroupSplit	XGBoost	0.523	42.156	0.511	116.905
	Ridge	0.288	47.194	0.208	145.595
	ElasticNet	0.211	49.658	0.052	159.212
	SVR	0.374	44.234	0.283	138.528
	ExtraTrees	0.466	40.875	0.513	114.195
	MLP	0.416	42.739	0.654	96.204

TABLE V: Benchmark performance on ΔVIIRS (dVIIRS).

Split	Model	R^2	RMSE	R^2_{tail}	$RMSE_{tail}$
RandomSplit	XGBoost	0.026	4.388	0.141	11.414
	Ridge	0.008	4.427	0.017	12.215
	ElasticNet	-0.000	4.446	-0.002	12.332
	SVR	-0.047	4.549	-0.014	12.403
	ExtraTrees	0.014	4.414	0.131	11.485
	MLP	0.010	4.424	0.020	12.193
GroupSplit	XGBoost	0.003	4.491	0.075	12.091
	Ridge	0.025	4.757	0.030	13.648
	ElasticNet	-0.000	4.817	-0.000	13.856
	SVR	-0.009	4.838	0.022	13.706
	ExtraTrees	0.093	4.587	0.168	12.639
	MLP	0.038	4.725	0.053	13.486

MoE: failure

Metric	**Base**	**MoE**	**Gain (%)**
Base (XGB) vs. MoE (2-expert)			
R^2_{all}	0.523	0.420	−19.64%
$RMSE_{all}$	42.156	46.474	−10.24%
R^2_{tail}	0.511	0.321	−37.06%
$RMSE_{tail}$	116.905	137.703	−17.79%

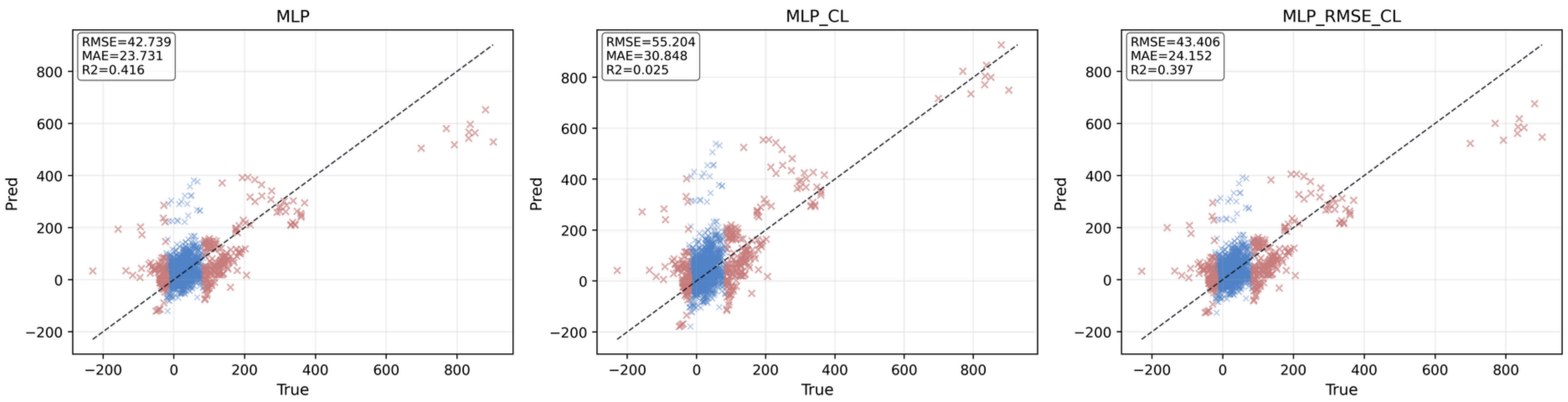
CL-calibrated

Metric	**Base (avg)**	**Gain (avg)**	**Gain (% avg)**
Mix (RMSE_CL) vs. Base			
R^2_{all}	0.378	+0.0045	+2.40%
$RMSE_{all}$	43.969	+0.143	+0.27%
R^2_{tail}	0.381	+0.0412	**+36.60%**
$RMSE_{tail}$	126.764	+4.081	**+2.93%**
CL-only vs. Base			
R^2_{all}	0.378	−0.3537	−124.66%
$RMSE_{all}$	43.969	−10.240	−22.20%
R^2_{tail}	0.381	+0.0500	**+44.94%**
$RMSE_{tail}$	126.764	+4.687	**+2.90%**

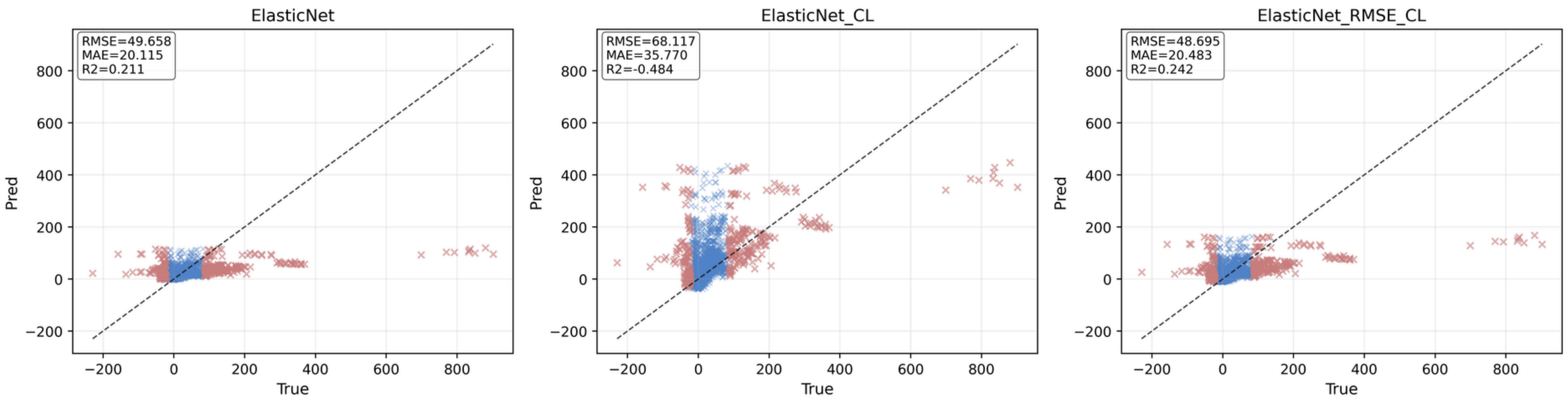
Results

Calibration mechanism

[GroupSplit(grid_id)] dWorldPop - MLP: Base vs CL vs RMSE_CL



[GroupSplit(grid_id)] dWorldPop - ElasticNet: Base vs CL vs RMSE_CL



Discussion & Conclusion

Key Conclusions

- Urban proxy dynamics are strongly heavy-tailed
- RMSE-anchored contrastive calibration:
  - Improves tail reliability
  - Preserves overall accuracy
- MoE is not robust under ambiguous tail routing

Practical Implication

When tail accuracy matters, anchoring is essential

References

[1] J. Ren, M. Zhang, C. Yu, and Z. Liu, "Balanced mse for imbalanced visual regression," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 7916–7925.

[2] Y. Yang, K. Zha, Y. Chen, H. Wang, and D. Katabi, "Delving into deep imbalanced regression," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 11 842–11 851.

[3] S. E. Yüksel, J. N. Wilson, and P. D. Gader, "Twenty years of mixture of experts," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 23, no. 8, pp. 1177–1193, 2012.

[4] I. C. Gormley and S. Frühwirth-Schnatter, "Mixtures of experts models," 2018, a chapter prepared for the forthcoming *Handbook of Mixture Analysis*.

[5] H. D. Nguyen and F. Chamroukhi, "Practical and theoretical aspects of mixture-of-experts modeling: An overview," *WIREs Data Mining and Knowledge Discovery*, vol. 8, p. e1246, 2018.

[6] D. Rügamer, F. Pfisterer, B. Bischl, and B. Grün, "Mixture of experts distributional regression: implementation using robust estimation with adaptive first-order methods," *AStA Advances in Statistical Analysis*, vol. 108, pp. 351–373, 2024.

[7] M. P. Paganelli, "Population as a gdp proxy in adam smith," SSRN working paper, 2021, forthcoming in *Journal of Scottish Philosophy*. [Online]. Available: https://ssrn.com/abstract=3826675